

Course Information

Course Number: ECCB 689
Course Title: Special Topics in Modeling Ecological Niches and Species' Distributions
Day: Tuesday, Thursday
Time: 2:50 – 4:05 pm
Location: WFES 411
Credit Hours: 3

Instructor Details

Instructor: Dr. Michelle Lawing
Office: 322 Wildlife, Fisheries, and Ecological Sciences Building (WFES)
E-Mail: alawing@tamu.edu
Office Hours: Tuesday 4:05 – 5:00 pm or by appointment

Course Description

This course examines ecological niche theory and species' geographic distributions through a project-centered, coding-based approach. These concepts provide a framework for building, interpreting, and evaluating ecological niche models, which are widely used in ecology, evolution, conservation, and disease ecology but rely on important assumptions and limitations. A central goal of the course is to develop the ability to recognize these assumptions and critically evaluate model-based inferences. The course is organized around structured R-based exercises that build toward an independent project. Each week, students will work with data to construct, evaluate, and interpret models while building reproducible code and outputs. Conceptual material is introduced and applied directly through coding and interpretation. Students should have some familiarity with R (e.g., importing data, basic functions, plotting), but support will be provided for those who need a refresher.

Course Prerequisites

Students in this class will have graduate student status or permission from the instructor to enroll.

Course Learning Outcomes

Upon completion of this course, students will be able to:

1. Explain and apply concepts of ecological niches and species' geographic distributions, including the relationship between environmental and geographic space.
2. Acquire, clean, and manage species occurrence and environmental data in R.
3. Build, evaluate, and compare ecological niche models using a range of modeling approaches.
4. Interpret model outputs and assess model assumptions, limitations, and sources of uncertainty.
5. Apply ecological niche models to ecological and conservation questions (e.g., biodiversity patterns, environmental change, invasions).

6. Develop and communicate a complete modeling workflow, including reproducible code, results, and interpretation.

Textbook and/or Resource Materials

Required

Peterson, A.T., Soberón, J., Pearson, R.G., Anderson, R.P., Martínez-Meyer, E., Nakamura, M. and Araújo, M.B., 2011. *Ecological niches and geographic distributions (MPB-49)* (Vol. 49). Princeton University Press. Freely available online:

https://www.researchgate.net/publication/230709994_Ecological_Niches_and_Geographic_Distributions

Lovelace, R., Nowosad, J. and Muenchow, J. 2019. *Geocomputation with R*. CRC Press. Freely available online: <https://r.geocompx.org/>

Suggested

Franklin, J., 2010. *Mapping species distributions: spatial inference and prediction*. Cambridge University Press.

Bivand, R.S., Pebesma, E.J., Gomez-Rubio, V. and Pebesma, E.J., 2013. *Applied spatial data analysis with R* (Vol. 2). New York: Springer.

Additional Materials

Readings will be drawn from primary literature and online resources and are selected to support weekly coding exercises and the term project.

Grading Policy - Grading

Grades will be based on a percentage of total points in the course and will follow a standard 90%, 80%, 70%, 60% delineation for letter grades.

Course grades are based on a combination of weekly applied assignments and a term project. Weekly assignments are designed to build components of the final project and will be evaluated based on code execution, output quality, and interpretation.

If you have concerns about a grade received on an individual assignment, please contact me within one week of receiving the grade. Grades will not be reconsidered at the end of the semester.

Major Due Dates	Assignment
9/11/26	Project topic
10/2/26	Data plan
10/23/26	Modeling plan
11/06/26	Preliminary results
12/1/26	Presentation
12/11/26	Final paper

Grading Policy Changes

Faculty must provide grading policies to students by the first class period. As such, faculty cannot change the course grading policy after the second class session. (See [Student Rule 10](#)).

Assignments Description

This course is organized around a series of structured, R-based assignments that build toward an independent project. Each week focuses on a core concept in ecological niche modeling that is introduced briefly and applied directly through coding, model construction, and interpretation using real data. The emphasis of the course is on understanding model assumptions, limitations, and inference through applied analysis.

Weekly assignments serve as the primary means of developing skills and will contribute directly to the final project. These assignments are due at the end of each week and will include code, output, and brief written interpretation.

The term project is an independent ecological niche modeling analysis developed incrementally over the semester. Students will select a problem, compile and process occurrence and environmental data, build and evaluate one or more models, and interpret the results in an ecological context. The project will demonstrate a complete, reproducible workflow, including code, model outputs, and written interpretation of results, assumptions, and limitations. The term project will be evaluated based on the quality of the modeling workflow, including data selection and processing, model construction and evaluation, clarity of interpretation, and reproducibility of code and outputs. The final submission will include a written report, figures, and annotated R code.

Section	Description	Grading
Weekly assignments	R-based exercises including data processing, model building, evaluation, and short interpretation responses	40 %
Term project	Topic, data plan, modeling plan, preliminary results, presentation, final paper	60 %
Total		100 %

Late Work Policy

Late work will be accepted for one week after the due date and will receive a deduction of ten percent of available points. Make up work for an excused absence is not considered late work and is exempted from the late work policy. (See Student Rule 7.)

Make up work for excused absences will be provided in accordance with university policy. Because assignments build sequentially and are completed in part through in-class coding and discussion, students are responsible for independently completing missed material. Make up assignments may differ from the original assignment but will assess the same learning objectives. Deadlines for make-up work will be established on a case-by-case basis in consultation with the instructor.

Course Schedule

Chapters under readings refer to Peterson *et al* (2011). During project development weeks (Weeks 10 – 14), students are expected to make measurable progress on their models, analyses, and written interpretation.

Week	Topic	Reading	Applied Assignment Focus
1	Foundations: what is a niche? environmental vs geographic space	Ch. 1–2	Set up R workflow; explore example dataset and outputs
2	Niches and distributions (fundamental vs realized)	Ch. 3	Visualize occurrence data and environmental space
3	Occurrence data: bias, sampling, cleaning	Ch. 4–5	Clean and filter occurrence data
4	Environmental data: scale, resolution, extent	Ch. 6–7 (part)	Extract and process environmental variables
5	Model structure and assumptions	Ch. 7–8	Prepare modeling dataset; select predictors
6	Algorithms I: GLMs, GAMs	Selected paper	Fit and interpret regression models
7	Algorithms II: Maxent, random forest	Selected paper	Fit and compare machine learning models
8	Model evaluation: discrimination vs calibration	Ch. 9	Evaluate model performance
9	Transferability and extrapolation	Selected paper	Test model transfer across space/time
10	Project development I: model refinement	—	Improve models; troubleshoot and refine inputs
11	Project development II: model performance	—	Compare models; select final approach
12	Interpretation: linking results to inference	—	Interpret outputs; identify assumptions and limitations
13	Revision: strengthen argument and evidence	—	Revise results and interpretation; improve figures
14	Synthesis: finalize workflow and narrative	—	Finalize code, outputs, and project structure
15	Presentations	—	Final project presentations
16	Finals Week	—	Final paper due

University Policies

Attendance Policy

The university views class attendance and participation as an individual student responsibility. Students are expected to attend class and to complete all assignments.

Please refer to [Student Rule 7](#) in its entirety for information about excused absences, including definitions, and related documentation and timelines.

Makeup Work Policy

Students will be excused from attending class on the day of a graded activity or when attendance contributes to a student's grade, for the reasons stated in Student Rule 7, or other reason deemed appropriate by the instructor.

Please refer to [Student Rule 7](#) in its entirety for information about makeup work, including definitions, and related documentation and timelines.

Absences related to Title IX of the Education Amendments of 1972 may necessitate a period of more than 30 days for make-up work, and the timeframe for make-up work should be agreed upon by the student and instructor" ([Student Rule 7, Section 7.4.1](#)).

"The instructor is under no obligation to provide an opportunity for the student to make up work missed because of an unexcused absence" ([Student Rule 7, Section 7.4.2](#)).

Students who request an excused absence are expected to uphold the Aggie Honor Code and Student Conduct Code. (See [Student Rule 24](#).)

Academic Integrity Statement and Policy

"An Aggie does not lie, cheat or steal, or tolerate those who do."

"Texas A&M University students are responsible for authenticating all work submitted to an instructor. If asked, students must be able to produce proof that the item submitted is indeed the work of that student. Students must keep appropriate records at all times. The inability to authenticate one's work, should the instructor request it, may be sufficient grounds to initiate an academic misconduct case" ([Section 20.1.2.3, Student Rule 20](#)).

You can learn more about the Aggie Honor System Office Rules and Procedures, academic integrity, and your rights and responsibilities at aggiehonor.tamu.edu.

Civil Rights, Free Speech, and Title IX Policies

Texas A&M University is committed to fostering a learning environment that is safe and productive for all. University policies and federal and state laws prohibit discrimination and harassment based on an individual's race, color, sex, (including pregnancy and related conditions), religion, national origin, age, disability, genetic information, veteran status, or any other legally protected characteristic. This includes forms of sex-based violence, such as sexual assault, sexual harassment, sexual exploitation, dating/domestic violence, and stalking.

Students can report discrimination/harassment, access supportive resources, or learn more about their options for resolving complaints on the [University's Civil Rights & Title IX webpage](#).

Students should be aware that all university employees (except medical or mental health providers) are mandatory reporters, which means that if they observe, experience or become aware of an incident that they reasonably believe to be discrimination/harassment alleged to have been committed by or against a person who was a student or employee at the time of the incident, the employee must report the incident to the university.

Americans with Disabilities Act (ADA) Policy

Texas A&M University is committed to providing equitable access to learning opportunities for all students. If you experience barriers to your education due to a disability or think you may have a disability, please contact the Disability Resources office on your campus (resources listed below). Disabilities may include, but are not limited to attentional, learning, mental health, sensory, physical, or chronic health conditions. All students are encouraged to discuss their disability related needs with Disability Resources and their instructors as soon as possible.

To request academic accommodations, contact the designated ADA office based on your location:

- Texas A&M University, College of Nursing, College of Dentistry, Irma Lerma Rangel College of Pharmacy College Station, College of Medicine, School of Public Health, Institute of Biosciences and Technology, EnMed Program, Bush School in Washington DC, Mays Business School – CityCentre, TAMU Engineering Academies, Texas A&M University Higher Education Center at McAllen and Texas A&M University at Galveston should contact [Disability Resources](#) at (979) 845-1637 or disability@tamu.edu.
- Texas A&M University School of Law should contact the Office of Student Affairs at (817) 212-4111 or law-disability@law.tamu.edu to request accommodations.
- Irma Lerma Rangel College of Pharmacy in Kingsville should contact the Disability Resource Center at Texas A&M University - Kingsville at (361) 593-3024 or drc.center@tamuk.edu to request accommodations.
- Texas A&M University College of Veterinary Medicine & Biomedical Sciences in Canyon should contact the Office of Student Accessibility at West Texas A&M University – Canyon at (806) 651-2335 or osa@wtamu.edu.
- Texas A&M University at Qatar (TAMUQ) should contact the campus psychologist, Dr. Steve Wilson +974-4423-0047 or stephen.wilson@qatar.tamu.edu.

If you are experiencing difficulties with your approved accommodations, contact the office responsible for approving your accommodations or the Texas A&M ADA Coordinator Julie Kuder at ADA.Coordinator@tamu.edu or (979) 458-8407.

Pregnancy Accommodations

Texas A&M provides reasonable accommodations to students due to pregnancy and/or related conditions, such as childbirth, recovery and lactation. Students should contact the University's [Pregnancy Coordinator](#) as soon as they become aware of the need for accommodation. Depending on the circumstances, accommodations could include extended time to complete assignments or exams, changes in course sequence, or modifications to the physical classroom environment. Texas A&M will also allow a voluntary leave of absence, ensure the availability of lactation space, and maintain grievance procedures to provide for the prompt and equitable resolution of complaints of sex discrimination. For information regarding pregnancy accommodations, email TIX.Pregnancy@tamu.edu.

Statement on Mental Health and Wellness

Texas A&M University recognizes that mental health and wellness are critical factors influencing a student's academic success and overall wellbeing. Students are encouraged to engage in healthy self-care practices by utilizing the resources and services available through [University Health Services](#) on its [mental health webpage](#). The [TELUS Health Student Support app](#) provides access to professional counseling in multiple languages anytime, anywhere by phone or chat, and the 988 Suicide & Crisis Lifeline offers 24-hour emergency support at 988 or [988lifeline.org](#).

Students needing a listening ear can contact University Health Services (979.458.4584) 24-hour emergency help is also available through the 988 Suicide & Crisis Lifeline (988) or at [988lifeline.org](#).

Statement on the Family Educational Rights and Privacy Act (FERPA)

FERPA is a federal law designed to protect the privacy of educational records by limiting access to these records, to establish the right of students to inspect and review their educational records and to provide guidelines for the correction of inaccurate and misleading data through informal and formal hearings. Currently enrolled students wishing to withhold any or all directory information items can do so within [howdy.tamu.edu](#) using the Directory Information Withholding Form. The complete [FERPA Notice to Students](#) and the student records policy is available on the Office of the Registrar webpage.

Items that can never be identified as public information are a student's social security number, citizenship, gender, grades, GPR or class schedule. All efforts will be made in this class to protect your privacy and to ensure confidential treatment of information associated with or generated by your participation in the class.

Directory items include name, UIN, local address, permanent address, email address, local telephone number, permanent telephone number, dates of attendance, program of study (college, major, campus), classification, previous institutions attended, degrees honors and awards received, participation in officially recognized activities and sports, medical residence location and medical residence specialization.

Optional Syllabus Statements

Free Speech and Civil Discourse

Texas A&M recognizes that the pursuit of truth through open and robust discourse is critical to academic inquiry. However, as a community of scholars, the university has an aspirational expectation that such discourse will be conducted in accordance with Aggie Core Values. In this "marketplace of ideas," we encourage civil dialogue creating an environment that allows individuals to express their ideas and to have their ideas challenged in respectful and responsible ways. Students can learn more about Freedom of Expression and Free Speech on the University's [website](#) about the [First Amendment](#).

Statement on use of Artificial Intelligence

According to the Texas A&M University Definitions of Academic Misconduct, plagiarism is the appropriation of another person's ideas, processes, results or words without giving appropriate credit ([aggiehonor.tamu.edu](#)). You should credit your use of anyone else's words, graphic images, or ideas using standard citation styles.

Artificial intelligence (AI) tools (e.g., ChatGPT, coding assistants) may be used in this course to support learning, including exploring ideas, troubleshooting code, and improving clarity of writing. These tools can be valuable for developing workflows and testing approaches; however, they must not replace your own reasoning, analysis, or interpretation.

All submitted work must reflect your understanding of the data, methods, and results. You are responsible for all code, outputs, and written interpretations you submit, including anything generated or modified with AI assistance. You should be able to explain and justify your analytical decisions and results independently.

Use of AI is appropriate for:

- debugging and refining code,
- generating example workflows or alternative approaches,
- clarifying concepts or improving written expression.

Use of AI is not appropriate for:

- generating complete assignment responses without substantive modification,
- producing interpretations or conclusions you do not understand,
- substituting for your own analysis of results.

If you use AI tools in your work, you should acknowledge this use briefly (e.g., in comments within code or a short note in your write-up). Failure to engage meaningfully with your work or to accurately represent your understanding may be considered a violation of academic integrity.